Learning When to Adapt

DISeL — Dynamic Input-Sensitive LoRA

Ali Zindari[1,2] **Xiaowen Jiang**[1,2] **Rotem Mulayoff**[2] **Sebastian U. Stich**[2]

[1]*Universität des Saarlandes* · [2]*CISPA Helmholtz Center for Information Security, Saarbrücken, Germany*

PROJECT github.com/alizindari/DISeL

CISPA HELMHOLTZ CENTER FOR INFORMATION SECURITY

PEFT · LLM FINE-TUNING · FORGETTING

TL;DR LoRA applies the *same* low-rank correction to *every* input. This forces an inherent compromise between adapting to the task and preserving pre-trained behavior. **DISeL** places lightweight input-dependent sigmoid gates on individual rank-one components, so the adapter is **active only when needed**. Same low-rank structure, near-zero added cost, and **retention close to the base model** across LLaMA, Mistral, and RoBERTa.

1 The forgetting problem

A fine-tuned model adds a learnable correction $f(\mathbf{x};\theta)$ on top of the frozen pre-trained weights:

$$\mathbf{y} \;=\; \mathbf{W}_0\,\mathbf{x} \;+\; f(\mathbf{x};\theta).$$

Full FT, LoRA, DoRA, AdaLoRA are all the special case $f(\mathbf{x};\theta) = \mathbf{\Delta}\,\mathbf{x}$, with a *fixed* matrix $\mathbf{\Delta}$ that does *not* depend on $\mathbf{x}$. They differ only in how $\mathbf{\Delta}$ is parameterized.

The same correction $\mathbf{\Delta}\,\mathbf{x}$ is therefore applied to fine-tuning inputs (where adaptation is needed) *and* to pre-training inputs (where the original mapping should be preserved).

Consequence: task accuracy improves, but the model loses general capability. This is ***catastrophic forgetting***.

> **Question.** *Is there a structural reason a fixed $\mathbf{\Delta}$ must forget, even in the simplest possible setting?*

2 A minimal linear analysis

Symmetric mixture of fine-tuning and pre-training populations:

$$\mathbf{x} \sim \begin{cases} p_{\text{ft}} & \text{w.p. } \tfrac{1}{2}, \\ p_{\text{pt}} & \text{w.p. } \tfrac{1}{2}, \end{cases} \qquad \mathbf{y} = \begin{cases} (\mathbf{W}_0 + \mathbf{M})\,\mathbf{x} & \text{when } \mathbf{x} \sim p_{\text{ft}}, \\ \mathbf{W}_0\,\mathbf{x} & \text{when } \mathbf{x} \sim p_{\text{pt}}. \end{cases}$$

Loss after subtracting the frozen mapping:

$$\mathcal{L}(f) = \tfrac{1}{2}\,\mathbb{E}_{p_{\text{ft}}}\big[\,\|f(\mathbf{x}) - \mathbf{M}\,\mathbf{x}\|^2\,\big] \;+\; \tfrac{1}{2}\,\mathbb{E}_{p_{\text{pt}}}\big[\,\|f(\mathbf{x})\|^2\,\big].$$

No fixed correction can do both

With equal second moments $\mathbf{\Sigma}$, the optimal *fixed* $\mathbf{\Delta}$ is

$$\mathbf{\Delta}^\star \;=\; \tfrac{1}{2}\,\mathbf{M}, \qquad \mathcal{L}(\mathbf{\Delta}^\star) \;=\; \tfrac{1}{4}\mathrm{Tr}\big(\mathbf{M}\,\mathbf{\Sigma}\,\mathbf{M}^\top\big) \;>\; 0.$$

No fixed $\mathbf{\Delta}$, including full FT, can both fit the FT signal *and* preserve pre-training: it applies only *half* of the desired update on p_{ft} and a non-zero spurious update on p_{pt}.

Bayes-optimal input-dependent correction

Allowing f to depend on $\mathbf{x}$, the pointwise minimizer is

$$f^\star(\mathbf{x}) \;=\; \pi_{\text{ft}}(\mathbf{x})\,\mathbf{M}\,\mathbf{x}, \qquad \pi_{\text{ft}}(\mathbf{x}) \;=\; \frac{p_{\text{ft}}(\mathbf{x})}{p_{\text{ft}}(\mathbf{x}) + p_{\text{pt}}(\mathbf{x})} \;\in\; [\,0,\,1\,],$$

$$\mathcal{L}_{\text{Bayes}} \;=\; \tfrac{1}{2}\int \frac{p_{\text{ft}}(\mathbf{x})\,p_{\text{pt}}(\mathbf{x})}{p_{\text{ft}}(\mathbf{x}) + p_{\text{pt}}(\mathbf{x})}\,\|\mathbf{M}\,\mathbf{x}\|^2\,d\mathbf{x}.$$

A *soft switch*: $\pi_{\text{ft}} \to 1$ on FT inputs, $\pi_{\text{ft}} \to 0$ on pre-training inputs. No measurable correction can do better.

Gaussian case: the sigmoid is exact

When $p_{\text{ft}} = \mathcal{N}(\boldsymbol{\mu}_{\text{ft}}, \mathbf{\Sigma})$ and $p_{\text{pt}} = \mathcal{N}(\boldsymbol{\mu}_{\text{pt}}, \mathbf{\Sigma})$ share covariance, the log-density ratio is *affine*, so the posterior is a sigmoid:

$$\log\frac{p_{\text{ft}}(\mathbf{x})}{p_{\text{pt}}(\mathbf{x})} = \mathbf{w}_g^\top\mathbf{x} + b_g, \qquad \pi_{\text{ft}}(\mathbf{x}) \;=\; \sigma\big(\mathbf{w}_g^\top\mathbf{x} + b_g\big),$$
$$\mathbf{w}_g \;=\; \mathbf{\Sigma}^{-1}(\boldsymbol{\mu}_{\text{ft}} - \boldsymbol{\mu}_{\text{pt}}) \in \mathbb{R}^d, \qquad b_g \;=\; \tfrac{1}{2}\big(\boldsymbol{\mu}_{\text{pt}}^\top\mathbf{\Sigma}^{-1}\boldsymbol{\mu}_{\text{pt}} - \boldsymbol{\mu}_{\text{ft}}^\top\mathbf{\Sigma}^{-1}\boldsymbol{\mu}_{\text{ft}}\big) \in \mathbb{R}.$$

Hence the Bayes-optimal correction has the closed form

$$f^\star(\mathbf{x}) = \sigma\big(\mathbf{w}_g^\top\mathbf{x} + b_g\big)\,\mathbf{M}\,\mathbf{x}.$$

DISeL realizes $f^\star$ exactly

Factor $\mathbf{M} = \mathbf{A}\mathbf{B}$ (which exists whenever $\mathrm{rank}(\mathbf{M}) \le r$). Set the gate parameters

$$\mathbf{W}_g \;=\; \mathbf{1}_r\,\mathbf{w}_g^\top \;\in\; \mathbb{R}^{r\times d_x}, \qquad \mathbf{b}_g \;=\; b_g\,\mathbf{1}_r \;\in\; \mathbb{R}^r.$$

Then every rank's gate carries the same value $g_i(\mathbf{x}) = \pi_{\text{ft}}(\mathbf{x})$, so

$$\mathbf{A}\,\mathbf{G}(\mathbf{x})\,\mathbf{B}\,\mathbf{x} = \pi_{\text{ft}}(\mathbf{x})\,\mathbf{A}\mathbf{B}\,\mathbf{x} = \pi_{\text{ft}}(\mathbf{x})\,\mathbf{M}\,\mathbf{x} = f^\star(\mathbf{x}).$$

A sigmoid-gated LoRA with the right parameters realizes the Bayes-optimal correction exactly. DISeL is the smallest learnable family containing this optimum.

3 Toy experiment

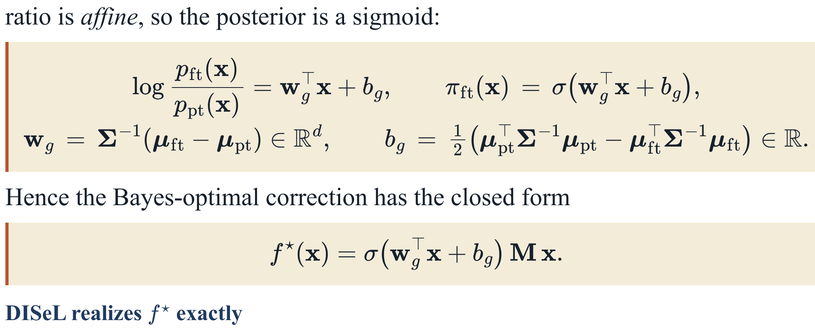


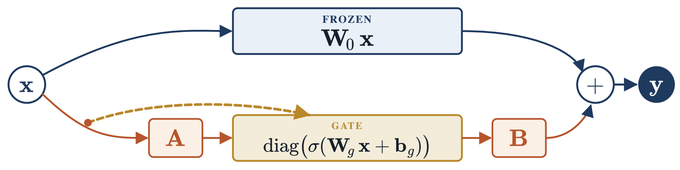
Fig. 1. *Left: LoRA and Full FT hit the fixed-correction floor; DISeL approaches the Bayes-optimal error on both domains. Right: learned gates concentrate near 1 on FT inputs and near 0 on pre-training inputs.*

4 DISeL: input-dependent gates over rank-1 components

We keep LoRA's low-rank structure $\mathbf{ABx}$ with $\mathbf{A} \in \mathbb{R}^{d_y\times r}$ and $\mathbf{B} \in \mathbb{R}^{r\times d_x}$, but multiply each rank-one component by a learnable input-dependent gate $g_i(\mathbf{x}) \in [0,1]$:

$$f(\mathbf{x};\theta) \;=\; \mathbf{A}\,\mathbf{G}(\mathbf{x})\,\mathbf{B}\,\mathbf{x} \;=\; \sum_{i=1}^{r} g_i(\mathbf{x})\,\mathbf{a}_i\,\mathbf{b}_i^\top\,\mathbf{x},$$
$$\mathbf{G}(\mathbf{x}) \;=\; \mathrm{diag}\big(g(\mathbf{x})\big), \qquad g(\mathbf{x}) \;=\; \sigma\big(\mathbf{W}_g\,\mathbf{x} + \mathbf{b}_g\big) \;\in\; [0,1]^r.$$

DISeL block



Design choices

- **Init.** $\mathbf{b}_g = -3\cdot\mathbf{1}$, so $\sigma(\mathbf{b}_g) \approx 0.05$. DISeL *starts equal to the pre-trained model.*
- **Sigmoid (not softmax).** Rank-one components are independent soft switches.
- **Two-LR.** Gates use a learning rate 5× larger than $(\mathbf{A},\mathbf{B})$, since they solve a *routing* task, not a fitting one.

5 LLaMA & Mistral on math & code

LLaMA-2-7B on *MetaMathQA*; Mistral-7B on *Magicoder-Evol-Instruct*. Retention = mean accuracy on 14 OOD benchmarks (commonsense, world knowledge, medical QA, …).

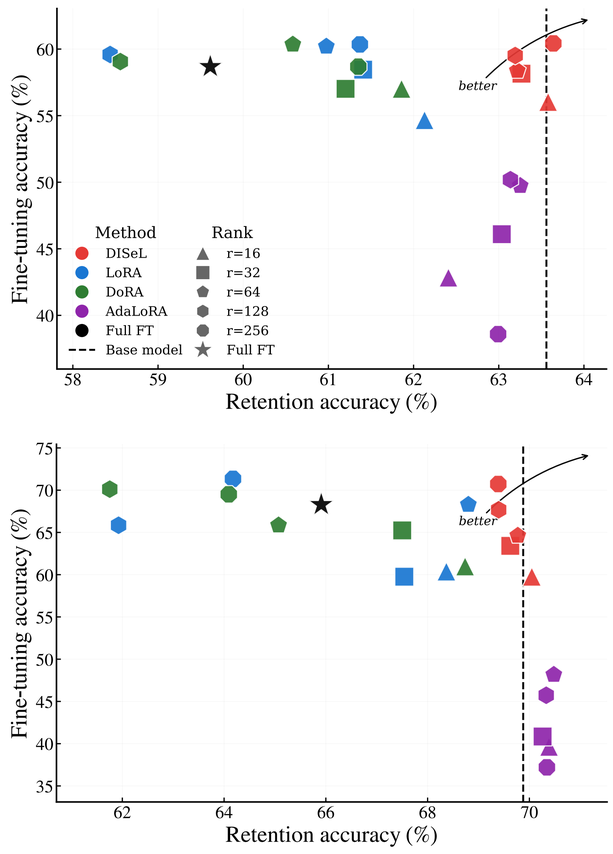


Fig. 2. *FT accuracy vs. retention for LLaMA / math (top) and Mistral / code (bottom). DISeL stays near the base across $r \in \{16, 32, 64, 128, 256\}$.*

- **DISeL**: ≈ 60% math accuracy (LLaMA) / > 70% code (Mistral) with retention within a fraction of the base.
- **LoRA / DoRA**: retention drops 4–8% at the largest ranks.
- **AdaLoRA**: keeps retention but FT accuracy collapses to 40–50%.
- **Implicit regularization.** Gates stabilize training at large LR / rank.

6 RoBERTa on GLUE

RoBERTa-base on five GLUE tasks, $r \in \{4, 8, 16, 32, 64\}$. Retention = mean MLM perplexity (base ≈ 6.4).

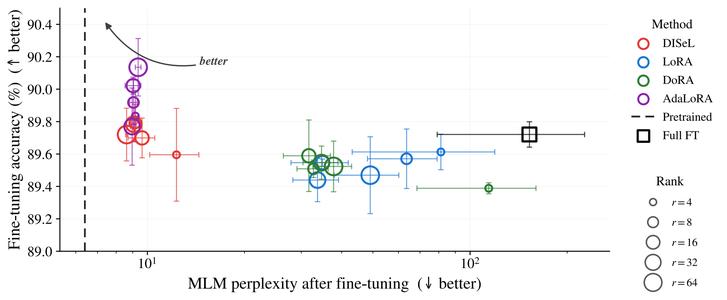


Fig. 3. *FT accuracy vs. MLM perplexity. DISeL & AdaLoRA stay near the pre-trained baseline; LoRA / DoRA forget more; Full FT forgets most.*

- **DISeL & AdaLoRA**: ppl ≈ 9, matching Full FT accuracy without any pruning schedule.
- **Full FT** tops accuracy but pushes ppl to ≈ 150; **LoRA / DoRA** sit at ppl 30–100.

7 Retention over training

Retention trajectory of LLaMA-2-7B on MetaMathQA across saved checkpoints, for ranks $r \in \{16, 32, 64, 128, 256\}$.

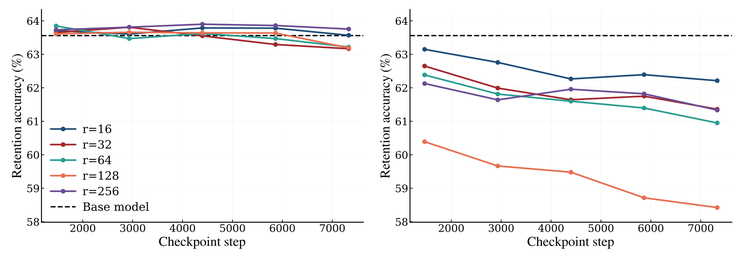


Fig. 6. **DISeL (left):** *near-flat retention at every rank, including the largest.* **LoRA (right):** *steady drift downward with both rank and training time.*

- **DISeL.** Retention within ≈ 0.5% of the base model at every rank and checkpoint.
- **LoRA.** Largest drift (≈ 5%) at $r = 128$, not $r = 256$: $r = 256$ needed a smaller learning rate (10^{-4}) for stability while $r = 128$ used a twice-larger step. *Rank alone does not decide forgetting — learning rate matters too.*
- **Mechanism.** DISeL gates stay inactive on pre-training inputs, so neither extra capacity nor larger learning rates leak into drift.

> **Rank decoupled from forgetting.** *DISeL safely allows larger r and larger learning rates with no retention penalty.*

8 Interpretability: *where & when* gates open

Held-out samples from three domains (math = FT, general text, code). Gate activations recorded at every adapted layer of the rank-32 LLaMA math adapter.

Domain-conditioned routing

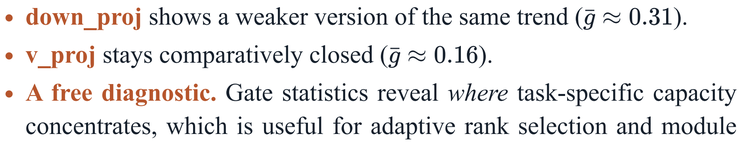


Fig. 4. *Early layers stay closed; middle and late layers separate cleanly: math (FT domain) opens gates much wider than general text (pre-training domain). Code, which shares structured reasoning with math, sits in between.*

- **Early layers** stay closed across all domains, so no adaptation is needed near the input.
- **Middle / late layers**: math $\bar{g} \approx 0.41 \to 0.41$; general text $\bar{g} \approx 0.18 \to 0.23$, a clear separation.
- **Code** sits between math and general text because it shares structured reasoning with math.

Module-level routing

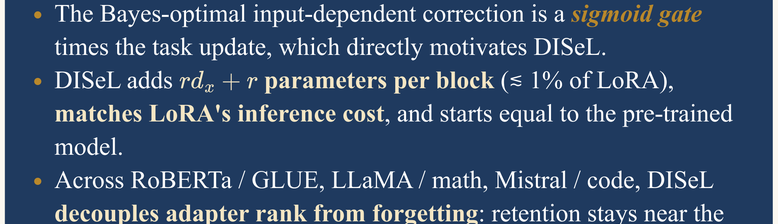


Fig. 5. *On math inputs, up_proj in middle/late layers carries the bulk of the adaptation (mean gate ≈ 0.57). v_proj remains comparatively closed.*

- **up_proj** dominates the math adaptation, with middle/late mean gate ≈ 0.57.
- **down_proj** shows a weaker version of the same trend ($\bar{g} \approx 0.31$).
- **v_proj** stays comparatively closed ($\bar{g} \approx 0.16$).
- **A free diagnostic.** Gate statistics reveal *where* task-specific capacity concentrates, which is useful for adaptive rank selection and module design.

Takeaways

- Any fixed adapter (full FT, LoRA, DoRA, AdaLoRA) must **compromise** between fitting the task and preserving pre-trained behavior. We prove this in a minimal linear-regression model and show it empirically.
- The Bayes-optimal input-dependent correction is a ***sigmoid gate*** times the task update, which directly motivates DISeL.
- DISeL adds $rd_x + r$ **parameters per block** (≲ 1% of LoRA), **matches LoRA's inference cost**, and starts equal to the pre-trained model.
- Across RoBERTa / GLUE, LLaMA / math, Mistral / code, DISeL **decouples adapter rank from forgetting**: retention stays near the base model while FT accuracy meets or beats LoRA, DoRA, AdaLoRA.
- Learned gate activations give a free **interpretability tool**: they reveal which layers and modules absorb task-specific signal (here: up_proj, middle/late layers).

Funded by the Helmholtz Association & ERC HORIZON #101170430 CollectiveMinds